

Clarkson Institute Summer School: Examples Sheet 9 - Reducibility to direct sums

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Complex vector spaces

1. Calculate the standard complex inner product of the vectors $(1, i, 1)$ and $(i, -1, 0)$ in $\mathbb{C}^3$.

Solution: We have:

$$(1, i, 1) \cdot (i, -1, 0) = i + i^*(-1) = 2i.$$

2. Show that:

$$(\mathbf{v}, \mathbf{w}) = \begin{pmatrix} v_1^* & v_2^* & v_3^* \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$$

defines an abstract inner product on $\mathbb{C}^3$. Evaluate the inner product of the vectors in Question 1 with respect to this inner product.

Solution: We have:

$$\left(\begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}, \begin{pmatrix} i \\ -1 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 1 & -i & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} i \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & -i & 1 \end{pmatrix} \begin{pmatrix} i \\ -2 \\ 0 \end{pmatrix} = 3i.$$

3. Apply the Gram-Schmidt procedure to produce an orthogonal basis from $\{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ for $\mathbb{C}^3$ with respect to the abstract inner product defined in Question 2.

Solution: Let $\mathbf{f}_1 = (1, 0, 0)$, $\mathbf{f}_2 = (1, 1, 0)$ and $\mathbf{f}_3 = (1, 1, 1)$. Then we have:

- $\mathbf{f}'_1 = \mathbf{f}_1 = (1, 0, 0)$.
- Next, we have:

$$\mathbf{f}'_2 = \mathbf{f}_2 - \frac{(\mathbf{f}_2, \mathbf{f}'_1)}{(\mathbf{f}'_1, \mathbf{f}'_1)} \mathbf{f}'_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

- Next, we subtract off any components in the directions of $\mathbf{f}'_1, \mathbf{f}'_2$. We have:

$$\begin{aligned}\mathbf{f}'_3 &= \mathbf{f}_3 - \frac{(\mathbf{f}_3, \mathbf{f}'_1)}{(\mathbf{f}'_1, \mathbf{f}'_1)} \mathbf{f}'_1 - \frac{(\mathbf{f}_3, \mathbf{f}'_2)}{(\mathbf{f}'_2, \mathbf{f}'_2)} \mathbf{f}'_2 \\ &= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{2}{2} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.\end{aligned}$$

4. Show that the matrix:

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix}$$

is unitary.

Solution: We have:

$$U^\dagger U = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix}^\dagger \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I,$$

as required.

Reducibility to direct sums

5. (a) Consider the left regular representation D of C_3 , comprising of three 3×3 matrices. Identify an invariant one-dimensional subspace and an invariant two-dimensional subspace of this representation.

Solution: A one-dimensional invariant subspace is V with basis $\{(1, 1, 1)\}$, as permuting the vector $(1, 1, 1)$ with any of the permutation matrices does nothing. Similarly a two-dimensional invariant subspace is V , the plane $z_1 + z_2 + z_3 = 0$. Permuting any vector (z_1, z_2, z_3) with this property gives another vector with this property, so it is invariant.

- (b) Extend the idea of your proof in part (a) to show that the left regular representation of any non-trivial finite group is always reducible.

Solution: A one-dimensional invariant subspace is $\{(1, 1, 1, \dots, 1)\}$ for any n -dimensional left regular representation; hence, this is always reducible.

6. Show that for $n \geq 3$, the representation D of D_n generated by:

$$D(r) = \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}, \quad D(m) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is irreducible.

Solution: The only invariant subspaces of $D(m)$ are generated by $(1, 0)$ and by $(0, 1)$ separately. But:

$$D(r) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos(2\pi/n) \\ \sin(2\pi/n) \end{pmatrix} \not\propto \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

unless $n = 1, 2$. Therefore, this representation is irreducible.

7. Let $D : G \rightarrow GL(2, \mathbb{C})$ be a two-dimensional reducible representation. Show that there exists a non-zero vector $\mathbf{v} \in \mathbb{C}^2$ such that:

$$D(g)\mathbf{v} = \lambda(g)\mathbf{v},$$

where $\lambda : G \rightarrow \mathbb{C}^*$ is a one-dimensional representation of G .

Solution: Since D is reducible, there is a one-dimensional invariant subspace V which must have a basis $\mathbf{v}$. We have by the invariant that:

$$D(g)\mathbf{v} = \lambda(g)\mathbf{v}$$

for some complex scalar $\lambda(g) \in \mathbb{C}$. We need to show that this is a one-dimensional representation. We have:

$$D(g)D(h)\mathbf{v} = \lambda(h)D(g)\mathbf{v} = \lambda(g)\lambda(h)\mathbf{v}, \quad D(gh)\mathbf{v} = \lambda(gh)\mathbf{v}.$$

Since these are the same, and $\mathbf{v} \neq \mathbf{0}$, we have $\lambda(gh) = \lambda(g)\lambda(h)$ as required.

We are almost there, but we need to show that $\lambda(g) \neq 0$. Suppose that $\lambda(g) = 0$ for some g . Then for any other element $h \in G$, we have:

$$\lambda(h) = \lambda(hgg^{-1}) = \lambda(h)\lambda(g)\lambda(g^{-1}) = 0.$$

Thus for all $h \in G$, we would have $\lambda(h) = 0$. This gives a non-zero vector $\mathbf{v}$ such that $D(g)\mathbf{v} = \mathbf{0}$, but this cannot occur because $D(g)$ is invertible so a bijection.

8. Let $D : G \rightarrow GL(n, \mathbb{C})$ be an irreducible representation, and let $d : G \rightarrow \mathbb{C}^*$ be a one-dimensional representation. Show that $D' : G \rightarrow GL(n, \mathbb{C})$ defined by $D'(g) = d(g)D(g)$ is irreducible.

Solution: Suppose for a contradiction that D' is reducible. Then there exists an invariant subspace V , such that $D'(g)\mathbf{v} \in V$ for all $\mathbf{v} \in V$. This gives:

$$d(g)D(g)\mathbf{v} \in V \quad \Rightarrow \quad D(g)\mathbf{v} \in V,$$

since $d(g) \in \mathbb{C}^*$, so we can divide by it and remain in the space. It follows that D is reducible, which is a contradiction.

9. Let D_1, D_2 be representations. Show that $D_1 \oplus D_2$ is equivalent to $D_2 \oplus D_1$.

Solution: Observe that:

$$(D_1 \oplus D_2)(g) = \begin{pmatrix} D_1(g) & 0 \\ 0 & D_2(g) \end{pmatrix}, \quad (D_2 \oplus D_1)(g) = \begin{pmatrix} D_2(g) & 0 \\ 0 & D_1(g) \end{pmatrix}.$$

The obvious intertwiner here is S given by:

$$S = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}.$$

Indeed, we have:

$$\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \begin{pmatrix} D_1(g) & 0 \\ 0 & D_2(g) \end{pmatrix} = \begin{pmatrix} 0 & D_2(g) \\ D_1(g) & 0 \end{pmatrix} = \begin{pmatrix} D_2(g) & 0 \\ 0 & D_1(g) \end{pmatrix} \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix},$$

so indeed they are equivalent.

Unitarity and Maschke's theorem

10. Consider the two-dimensional representation D of D_2 generated by:

$$D(r) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D(m) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Identify two invariant subspaces of the representation. Show that it is completely reducible to a direct sum of one-dimensional representations, which you should identify.

Solution: Two invariant subspaces are V_+ with basis $(1, 0)$ and V_- with basis $(0, 1)$. Observe that:

$$D(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D(r) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D(m) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D(rm) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

so this is obviously a direct sum $D_1 \oplus D_2$, given by:

$$D_1(e) = 1, D_1(r) = -1, D_1(m) = 1, D_1(rm) = -1,$$

$$D_2(e) = 1, D_2(r) = -1, D_2(m) = -1, D_2(rm) = 1.$$

11. Consider the two-dimensional representation D of C_n generated by:

$$D(r) = \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}.$$

(a) Show that this representation is unitary with respect to the standard complex scalar product.

Solution: We have:

$$D(r)^\dagger = \begin{pmatrix} \cos(2\pi/n) & \sin(2\pi/n) \\ -\sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix},$$

which is a clockwise rotation by $2\pi/n$ about the origin, and hence $D(r)^\dagger = D(r)^{-1}$.

(b) Show that there are two invariant subspaces V_+, V_- with $\{(1, i)\}$ a basis for V_+ and $\{(1, -i)\}$ a basis for V_- . Show that these are mutually orthogonal subspaces with respect to the standard complex scalar product on $\mathbb{C}^2$.

Solution: Checking the invariance, we have:

$$\begin{aligned} D(r) \begin{pmatrix} 1 \\ \pm i \end{pmatrix} &= \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix} \begin{pmatrix} 1 \\ \pm i \end{pmatrix} \\ &= \begin{pmatrix} \cos(2\pi/n) \mp i \sin(2\pi/n) \\ \sin(2\pi/n) \pm i \cos(2\pi/n) \end{pmatrix} \\ &= \begin{pmatrix} \cos(2\pi/n) \mp i \sin(2\pi/n) \\ \pm i(\cos(2\pi/n) \mp i \sin(2\pi/n)) \end{pmatrix} \\ &= e^{\mp 2\pi i/n} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}. \end{aligned}$$

These subspaces are mutually orthogonal, since $(1, i) \cdot (1, -i) = 1 + (i)^*(-i) = 1 - 1 = 0$.

(c) Find an intertwining matrix S such that $S^{-1}D(r)S$ is block diagonal. Hence, show that $D \cong D_+ \oplus D_-$, where $D_+(r) = e^{2\pi i/n}$ and $D_-(r) = e^{-2\pi i/n}$ are one-dimensional irreducible representations.

Solution: We should take:

$$S = \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \quad \Rightarrow \quad S^{-1} = -\frac{1}{2i} \begin{pmatrix} -i & -1 \\ -i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$

Then we have:

$$\begin{aligned} S^{-1}D(r)S &= \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} \cos(2\pi/n) & \sin(2\pi/n) \\ -\sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} e^{2\pi i/n} & e^{-2\pi i/n} \\ ie^{2\pi i/n} & -ie^{-2\pi i/n} \end{pmatrix} \\ &= \begin{pmatrix} e^{2\pi i/n} & 0 \\ 0 & e^{-2\pi i/n} \end{pmatrix}. \end{aligned}$$

Thus indeed D is reducible to the sum $D_1 \oplus D_2$, where $D_1(r) = e^{2\pi i/n}$ and $D_2(r) = e^{-2\pi i/n}$.

12. Consider the two-dimensional representation D of C_2 generated by:

$$D(r) = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}.$$

(a) Show that this indeed extends to a representation of C_2 .

Solution: We check:

$$D(r)^2 = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I,$$

so all relations are satisfied.

(b) Show that D is not unitary with respect to the standard complex scalar product.

Solution: We have:

$$D(r)^\dagger = \begin{pmatrix} 1 & 0 \\ -2 & -1 \end{pmatrix}, \quad D(r)^{-1} = D(r),$$

using $D(r)^2 = I$. Note $D(r)^\dagger \neq D(r)$.

(c) Construct a new abstract complex scalar product $(\cdot, \cdot)$, as per the proof of Maschke's theorem, and show that it is given by:

$$(\mathbf{v}, \mathbf{w}) = \begin{pmatrix} v_1^* & v_2^* \end{pmatrix} \begin{pmatrix} 2 & -2 \\ -2 & 6 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}.$$

Solution: We take the abstract inner product:

$$[\mathbf{v}, \mathbf{w}] = \sum_{g \in C_2} D(g)\mathbf{v} \cdot D(g)\mathbf{w},$$

where $\cdot$ is the usual complex scalar product. We therefore have:

$$\begin{aligned} [\mathbf{v}, \mathbf{w}] &= \mathbf{v} \cdot \mathbf{w} + \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix} \mathbf{v} \cdot \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix} \mathbf{w} \\ &= v_1^* w_1 + v_2^* w_2 + \begin{pmatrix} v_1 - 2v_2 \\ -v_2 \end{pmatrix} \cdot \begin{pmatrix} w_1 - 2w_2 \\ -w_2 \end{pmatrix} \\ &= v_1^* w_1 + v_2^* w_2 + (v_1 - 2v_2)^* (w_1 - 2w_2) + v_2^* w_2 \\ &= 2v_1^* w_2 - 2v_1^* w_2 - 2v_2^* w_1 + 6v_2^* w_2 \\ &= \begin{pmatrix} v_1^* & v_2^* \end{pmatrix} \begin{pmatrix} 2 & -2 \\ -2 & 6 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}. \end{aligned}$$

- (d) Show that V_+, V_- are invariant subspaces, where V_+ has basis $\{(1, 0)\}$ and V_- has basis $\{(1, 1)\}$. Verify that these are mutually orthogonal with respect to the abstract complex scalar product defined above.

Solution: Note that:

$$D(r) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad D(r) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}.$$

These are mutually orthogonal, since:

$$\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right] = -2 + 2 = 0.$$

- (e) Construct an intertwining matrix that transforms D to block diagonal form. Hence decompose D into a direct sum of irreducible one-dimensional representations.

Solution: The intertwining matrix is:

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad S^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Then we have:

$$\begin{aligned} S^{-1}D(r)S &= \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

Thus $D = D_1 \oplus D_2$ where D_1 is the trivial representation, and D_2 is the representation $D_2(e) = 1, D_2(r) = -1$.